

10,000 real PII-free documents: a different labelling standard, and it hurts

Results

10,000 real govdocs documents asserted PII-free were tested against the document gate. **They cannot be used as they stand**, for two independent reasons found before any training, and a third found by measuring.

Admissibility — 2,874 of 10,000 disqualified before any use

	n	why
in a sealed evaluation corpus	587	training on them leaks the measurement into the model
label conflict with existing gold	782	existing dual-judge says PII-bearing (sensitivity: low); this manifest says none
overlapping any existing corpus	2,874	already labelled; re-adding is duplication at best
usable — no existing gold at all	7,126	what the rest of this report uses

The conflict rate on the overlap is **27.2%**. That is not a rounding difference; it is two labelling passes disagreeing about the same files — 002/002518.doc (Budget Proposal), 010/010044.txt (Audit Report) and 780 others.

They are distinguishable from the corpus they would join

comparison	AUC
govdocs2 train vs govdocs2 eval negatives (<i>control: same corpus, two halves</i>)	0.5376
new clean docs vs govdocs2 eval negatives	0.7631
new clean docs vs govdocs2 train negatives	0.8087
<i>govdocs2 vs datax negatives (control: genuinely different corpora)</i>	<i>0.9858</i>

Two halves of the existing judged-clean govdocs2 documents are effectively indistinguishable (0.54). The new documents separate from them at **0.76–0.81**, despite all being govdocs. Something systematic differs, and the 27.2% conflict rate names it: the bar for "clean" is not the same one.

The gate agrees. It wrongly fires on **20.1%** of the new documents against **13.1%** of the sealed real negatives — they carry more PII-like material.

Adding them makes the model worse

Balanced gate, 5,700 documents added to training, 1,426 held out, scored on the sealed real corpora:

clean-doc weight	AUC real	AUC datax	AUC govdocs2	recall @ spec 0.95	best recall @ bars	held-out fire
0.0 (baseline)	0.8804	0.8888	0.8746	0.5662	0.6231	9.0%
1.0	0.8764	0.8864	0.8706	0.5475	0.6020	5.3%
4.0	0.8744	0.8858	0.8687	0.5501	0.5987	4.8%
12.0	0.8721	0.8841	0.8664	0.5438	0.5905	4.8%

Every column degrades monotonically with weight. This is worse than the synthetic corpus, which was merely flat: real documents labelled to a different standard actively teach the gate to suppress the signals the sealed gold rewards.

TL;DR

- **587 documents are in the sealed eval set.** Excluded — training on them would have invalidated every number in this project.
- **782 have conflicting gold**, 27.2% of the overlap. The existing dual-judge process calls them PII-bearing; this manifest calls them clean.
- **The usable 7,126 are distinguishable from existing judged-clean govdocs2 at AUC 0.76–0.81**, against a same-corpus control of 0.54.
- **Adding them degrades the gate monotonically** — sealed real AUC 0.8804 → 0.8721, best constrained recall 0.6231 → 0.5905.
- **Do not use them as negatives at all.** Adjudicated: the existing gold is right on ~97% of the 782 conflicts, so the corpus is mislabelled, not merely labelled to a different standard.
- The corpus itself is fine and well-formed; the problem is that its "clean" means something different from the corpus the model is measured against.

Date	2026-08-26
Project	projects/pii-head-to-head-v1
Scope	document gate only; 7,126 admissible documents of 10,000
Outcome	rejected for training; labelling-standard conflict identified

Model scoreboard

Model	Dataset	F1@ 5	F1@ 3	F1@ 1	Macro F1	Docs/sec	ms/doc
fusion-1k	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
fusion-1k	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
fusion-1k	20000_pii_holdout_20.00k				0.477	1360.8	0.7
fusion-1k	30000_pii2_eval_25.15k				0.413	1360.8	0.7
fusion-1k	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7
fusion-1k	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7
fusion-1k	5617_nemotron_eval_5.36k					1360.8	0.7
fusion-1k	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
steady-cascade	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.266	314.1	3.2
steady-cascade	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.729	314.1	3.2
steady-cascade	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.709	314.1	3.2
steady-cascade	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.757	314.1	3.2
steady-cascade	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.657	314.1	3.2
steady-cascade	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
steady-cascade	5617_nemotron_eval_5.36k					314.1	3.2
steady-cascade	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
fusion-12k	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
fusion-12k	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
fusion-12k	20000_pii_holdout_20.00k				0.485	270.3	3.7
fusion-12k	30000_pii2_eval_25.15k				0.390	270.3	3.7
fusion-12k	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
fusion-12k	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
fusion-12k	5617_nemotron_eval_5.36k					270.3	3.7
fusion-12k	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7
steady-cascade-bal	10360_betterdataai_ner_silver_eval_10.36k	0.208	0.174	0.126	0.289	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
steady-cascade-bal	10626_ai4privacy_pii_masking_eval_10.63k	0.748	0.693	0.593	0.742	314.1	3.2
steady-cascade-bal	20000_pii_holdout_20.00k	0.942	0.927	0.886	0.704	314.1	3.2
steady-cascade-bal	30000_pii2_eval_25.15k	0.890	0.852	0.768	0.757	314.1	3.2
steady-cascade-bal	38937_openpii_pii_eval_38.94k	0.988	0.986	0.977	0.658	314.1	3.2
steady-cascade-bal	4000_datax-dualjudge-evalset-1.32k	0.572	0.503	0.329		314.1	3.2
steady-cascade-bal	5617_nemotron_eval_5.36k					314.1	3.2
steady-cascade-bal	6589_govdocs2-dualjudge-eval20-3.53k	0.571	0.546	0.475		314.1	3.2
cascade_f65b00	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.247	314.1	3.2
cascade_f65b00	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.710	314.1	3.2
cascade_f65b00	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.728	314.1	3.2
cascade_f65b00	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.765	314.1	3.2
cascade_f65b00	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.626	314.1	3.2
cascade_f65b00	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f65b00	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f65b00	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f65b10	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.247	314.1	3.2
cascade_f65b10	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.711	314.1	3.2
cascade_f65b10	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.735	314.1	3.2
cascade_f65b10	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.786	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_f65b10	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.634	314.1	3.2
cascade_f65b10	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f65b10	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f65b10	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f65b25	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.247	314.1	3.2
cascade_f65b25	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.722	314.1	3.2
cascade_f65b25	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.743	314.1	3.2
cascade_f65b25	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.786	314.1	3.2
cascade_f65b25	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.647	314.1	3.2
cascade_f65b25	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f65b25	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f65b25	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f65b50	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.282	314.1	3.2
cascade_f65b50	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.742	314.1	3.2
cascade_f65b50	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.754	314.1	3.2
cascade_f65b50	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.786	314.1	3.2
cascade_f65b50	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.663	314.1	3.2
cascade_f65b50	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f65b50	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f65b50	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_f70b00	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.274	314.1	3.2
cascade_f70b00	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.707	314.1	3.2
cascade_f70b00	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.713	314.1	3.2
cascade_f70b00	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.761	314.1	3.2
cascade_f70b00	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.638	314.1	3.2
cascade_f70b00	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f70b00	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f70b00	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f70b10	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.274	314.1	3.2
cascade_f70b10	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.710	314.1	3.2
cascade_f70b10	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.720	314.1	3.2
cascade_f70b10	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.776	314.1	3.2
cascade_f70b10	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.641	314.1	3.2
cascade_f70b10	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f70b10	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f70b10	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f70b25	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.274	314.1	3.2
cascade_f70b25	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.714	314.1	3.2
cascade_f70b25	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.724	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_f70b25	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.776	314.1	3.2
cascade_f70b25	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.647	314.1	3.2
cascade_f70b25	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f70b25	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f70b25	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f70b50	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.273	314.1	3.2
cascade_f70b50	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.732	314.1	3.2
cascade_f70b50	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.733	314.1	3.2
cascade_f70b50	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.776	314.1	3.2
cascade_f70b50	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.660	314.1	3.2
cascade_f70b50	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f70b50	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f70b50	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f75b00	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.266	314.1	3.2
cascade_f75b00	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.716	314.1	3.2
cascade_f75b00	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.696	314.1	3.2
cascade_f75b00	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.748	314.1	3.2
cascade_f75b00	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.645	314.1	3.2
cascade_f75b00	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f75b00	5617_nemotron_eval_5.36k					314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_f75b00	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f75b10	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.266	314.1	3.2
cascade_f75b10	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.716	314.1	3.2
cascade_f75b10	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.700	314.1	3.2
cascade_f75b10	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.758	314.1	3.2
cascade_f75b10	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.647	314.1	3.2
cascade_f75b10	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f75b10	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f75b10	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f75b25	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.266	314.1	3.2
cascade_f75b25	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.721	314.1	3.2
cascade_f75b25	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.702	314.1	3.2
cascade_f75b25	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.758	314.1	3.2
cascade_f75b25	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.650	314.1	3.2
cascade_f75b25	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f75b25	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f75b25	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2
cascade_f75b50	10360_betterdataai_ner_silver_eval_10.36k	0.210	0.177	0.129	0.266	314.1	3.2
cascade_f75b50	10626_ai4privacy_pii_masking_eval_10.63k	0.742	0.692	0.595	0.729	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
cascade_f75b50	20000_pii_holdout_20.00k	0.940	0.925	0.884	0.709	314.1	3.2
cascade_f75b50	30000_pii2_eval_25.15k	0.889	0.851	0.769	0.757	314.1	3.2
cascade_f75b50	38937_openpii_pii_eval_38.94k	0.981	0.979	0.971	0.657	314.1	3.2
cascade_f75b50	4000_datax-dualjudge-evalset-1.32k	0.557	0.484	0.319		314.1	3.2
cascade_f75b50	5617_nemotron_eval_5.36k					314.1	3.2
cascade_f75b50	6589_govdocs2-dualjudge-eval20-3.53k	0.559	0.535	0.470		314.1	3.2

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.